Granger Causality Summary — Part 2

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At a Glance

Granger causality is a **forecasting test** that asks "does the past of X make the forecast of Y more accurate?"; it does not prove a true cause-and-effect relationship. The conclusions of Part 2 come down to three points.

1. **Granger causality is a forecasting test.** An F-test compares a model that forecasts Y from its own past only (restricted) with one that adds the past of X (unrestricted). The two directions, X→Y and Y→X, are tested separately.
2. **It is not true causation.** A hidden common driver (a confounder), market expectations or release lags alone can create a forecasting relationship. Always read the result only as "X helps forecast Y".
3. **Used honestly, it becomes a map of information flow.** Validated out of sample and with related variables controlled for, it shows which series carries information earlier than another (lead–lag).

1. The Core Question and the Definition

X Granger-causes Y means that the forecast error of Y is **clearly smaller** when X's past is used as well than when only Y's own past is used. Nothing more, nothing less.

The question that follows from Part 1

The VAR in Part 1 estimated off-diagonal coefficients (cross effects) linking each series to the past of the others. Now we ask whether those links are statistically real.

- First, forecast Y as well as possible from its own past alone.
- Then add the past of X. Does the forecast improve?
- If it does, X contains information about Y's future. That is, X **leads** Y in the forecasting sense.

The idea was proposed by Clive Granger (1969), who received the Nobel Prize in Economics in 2003. Because it compares only the errors of two forecasts without assuming any theory, it translates directly into a standard statistical test.

Definition, in terms of forecast error

MSE (mean squared error) measures how far a forecast misses on average. Y_{<t} means "all of Y before t".

$$X \text{ Granger-causes } Y \iff \mathrm{MSE}(Y_t \mid Y_{<t}, X_{<t}) < \mathrm{MSE}(Y_t \mid Y_{<t})$$

- The left side is the forecast error using the past of both series; the right side is the forecast error using only Y's past.
- If adding X reduces the error **strictly**, X Granger-causes Y.
- Here 'cause' only means "helps to forecast". The name is a historical convention and, as Section 4 shows, it invites misunderstanding.

Rewritten with VAR coefficients

Inside a VAR, Granger causality becomes a question of whether certain off-diagonal coefficients are all 0. In a two-variable VAR(p), the lag coefficient of Y2 in the Y1 equation is a12 (a12^(i) for each lag i).

$$\text{Bivariate VAR}(p): \quad Y_2 \not\Rightarrow Y_1 \iff a_{12}^{(1)} = a_{12}^{(2)} = \cdots = a_{12}^{(p)} = 0$$

In other words, "Y2 does not Granger-cause Y1" is the same statement as "in the Y1 equation, the coefficients on Y2's past are all 0 from lag 1 to p". It tests exactly the coefficients that Part 1 called "the heart of the VAR": the cross terms.

2. The Test: Restricted vs Unrestricted Model

The test is an F-test comparing the residuals of two regressions. If adding the lags of X reduces the residuals by more than chance can plausibly explain, we conclude that X Granger-causes Y.

Two regressions

$$\text{Restricted (R):} \quad Y_t = c + \sum_{i=1}^{p} a_i \, Y_{t-i} + u_t$$

$$\text{비제약 모형 (U):} \quad Y_t = c + \sum_{i=1}^{p} a_i \, Y_{t-i} + \sum_{i=1}^{p} b_i \, X_{t-i} + \varepsilon_t$$

The restricted model imposes the restriction "the coefficients b on X are all 0". The unrestricted model lifts that restriction. The null hypothesis is therefore as follows.

$$H_0: \; b_1 = b_2 = \cdots = b_p = 0 \qquad \text{(X's past adds nothing to forecasting Y)}$$

The lags are tested **together (jointly)** rather than one at a time, because the individual b may look small yet be significant taken together.

F-statistic

RSS (residual sum of squares) is the size of the error the model fails to explain. Adding variables always lowers RSS, so the F-statistic judges whether "the drop is more than chance".

$$F = \frac{(\mathrm{RSS}_R - \mathrm{RSS}_U)/q}{\mathrm{RSS}_U/(T-k)} \sim F(q, T-k)$$

Symbol	Meaning
RSS_R, RSS_U	Residual sums of squares of the restricted and unrestricted models. Always RSS_U ≤ RSS_R
q	Number of restrictions = number of X lags set to 0 (usually p)
T	Number of observations
k	Number of estimated coefficients in the unrestricted model (1 constant + p lags of Y + p lags of X)

The numerator is "the error removed per lag of X", and the denominator is "the remaining noise per degree of freedom". If the numerator is sufficiently larger than the denominator, X contributes more than noise. If F exceeds the 5% critical value of the F(q, T−k) distribution, reject the null hypothesis.

Example result: testing whether a sector index (X) forecasts an individual stock (Y) with 2 lags and 100 observations, RSS fell from 52 to 45 and F = 7.39. This is well above the 5% critical value of 3.09, so "the sector Granger-causes the stock". The calculation is in the practice materials.

Decision

- **Reject**: X's past contributes to forecasting Y → X Granger-causes Y.
- **Fail to reject**: X adds nothing beyond Y's own past.

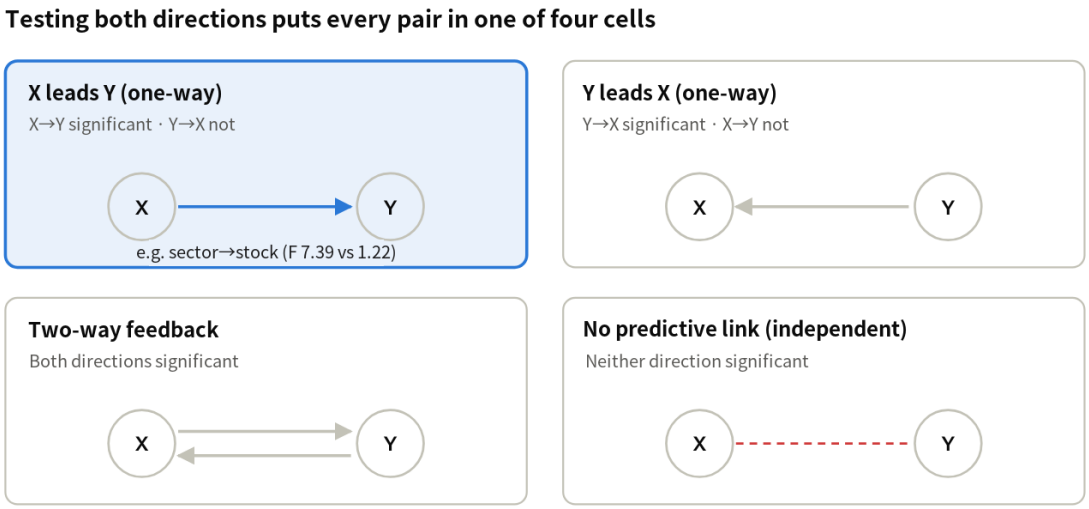
Five-step procedure

1. **Make stationary**: difference X and Y into stationary series (Part 1). With non-stationary series the F-distribution does not apply, and spurious significance results.
2. **Choose the lag**: use the VAR's lag order p as it is, with the same p for both variables.
3. **Estimate both models**: fit the restricted and the unrestricted model, each by OLS.
4. **F-test**: jointly test whether X's lag coefficients are all 0.
5. **Reverse test**: swap the roles of X and Y and test again. X→Y and Y→X are **separate questions** and may have different answers.

Continuing the example, the reverse direction (does the stock forecast the sector?) gave F = 1.22, below 3.09 and not significant. The pair is therefore "one-way, X→Y".

3. Four Outcomes

When the two directions are tested separately, each is either significant or not. The combination of the two answers places every pair in one of four cells.



The four outcomes of the Granger test

- **One-way (X→Y or Y→X)**: there is a clear leading order in the forecasting sense. The sector–stock example above belongs here.
- **Bidirectional feedback**: each series' past forecasts the other. It appears in pairs that influence each other, such as interest rates and stock prices.
- **Independent**: neither adds anything to forecasting the other. Same-period links (instantaneous causality, Section 5) may still remain.

4. The Most Important Warning: Granger Causality Is Not True Causation

"X Granger-causes Y" means "X's past forecasts Y", not "X makes Y happen". Because the name promises far more than the test delivers, generations of researchers have misread it.

What it proves	What it cannot prove
X's past contains information about Y's future	Changing X (intervening) changes Y
A directional, time-ordered forecasting relationship	Some causal mechanism lies behind it
A signal usable for forecasting and screening candidate variables	No hidden third variable moves both together

Three ways forecasting and causation part company

1. **Confounder**: a hidden common driver Z moves X and Y with different lags.

2. **Expectations**: markets price in the future ahead of time, so the apparent lead–lag can run opposite to the true causal direction. For example, stock prices "Granger-cause" the business cycle, yet stock prices do not create the cycle; they merely reflect the economic outlook first.
3. **Timing illusion**: because of the sampling interval or reporting delays, events that happened at the same time are observed with a lag.

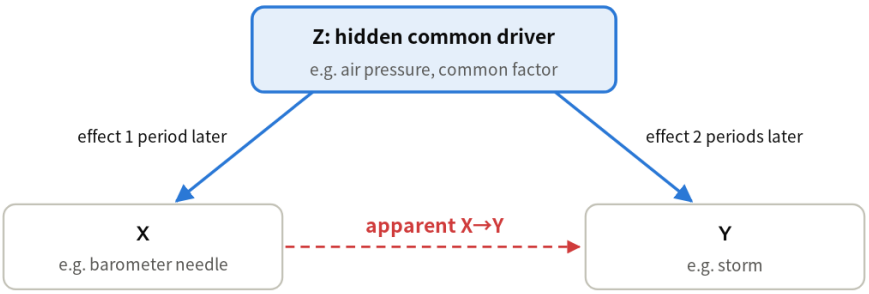
How a confounder creates a false link

Take the simplest example. Suppose Z affects X one period later and Y two periods later.

$$X_t = Z_{t-1} + u_t, \qquad Y_t = Z_{t-2} + v_t \quad \Longrightarrow \quad X_{t-1} \approx Z_{t-2} \text{ predicts } Y_t$$

There is no link at all between X and Y. Yet X_{t−1} effectively contains Z_{t−2}, so it forecasts Y_t well. The test will strongly reject in favour of "X→Y". In the course simulation, ignoring Z gave F = 1735.8, but once Z was controlled for in the regression, F ≈ 0 and the link vanished.

A hidden common driver Z creates a spurious X→Y forecasting link



Ignoring Z: F = 1735.8 (strongly significant) · controlling for Z: F ≈ 0 (vanishes)

False Granger causality created by a confounder

Solid lines are actual influences; dashed lines are the apparent links the test found. Because X reacts to Z sooner, X's past forecasts Y, yet there is no mechanism at all between X and Y.

The barometer story

When the barometer needle drops, a storm soon follows. The barometer's past "Granger-causes" the storm. But breaking the barometer will not stop the storm. The hidden cause, air pressure (Z), moves both, and the barometer simply reacts first. The test correctly found forecasting information, and nothing more.

The same holds in finance. One asset may lead another because both follow the same factor and one reflects common news faster. Treating this as causation and trading on it fails the moment the factor structure changes.

How to read it honestly

- "X Granger-causes Y" = "X's past forecasts Y". Stop there.
- Mentally add "in a purely statistical, non-causal sense".
- Make causal claims only when you have an experiment or a careful identification design.

5. Instantaneous Causality and Block Exogeneity

Granger causality deals only with **lagged** forecasting. It is extended to same-period links through error correlation (instantaneous causality) and, with three or more variables, to groups of variables (block exogeneity).

Instantaneous causality

When the shocks of two series move together within the same period, the link shows up not in the coefficient matrix A but in the covariance of the VAR errors.

$$\text{Instantaneous causality} \iff \mathrm{Cov}(\varepsilon_{1,t},\, \varepsilon_{2,t}) \neq 0$$

- It means that the off-diagonal element σ12 of the residual covariance matrix Σ from Part 1 is not 0.
- **It has no direction.** Since Cov(ε1, ε2) = Cov(ε2, ε1), the data alone cannot tell which comes first.
- It is precisely this property that leads Part 3 to orthogonalize the shocks, at which point an assumption about the ordering of the variables becomes necessary.

Beyond pairs: block exogeneity

With three or more variables, testing X→Y on its own is easily fooled by a third variable (the confounder of Section 4). So we test within a VAR that includes the other variables, or jointly test whether one **group (block)** of variables forecasts another. Splitting the variable vector into two blocks Y1 and Y2, the null hypothesis is as follows.

$$\mathbf{Y}_t = \begin{pmatrix} \mathbf{Y}_{1,t} \\ \mathbf{Y}_{2,t} \end{pmatrix}, \qquad H_0 :\ A_{12}^{(1)} = A_{12}^{(2)} = \cdots = A_{12}^{(p)} = \mathbf{0}$$

- A12^(i) is the submatrix of the i-th lag coefficient matrix that collects only "the Y2 coefficients in the Y1 equations".
- If these blocks are all 0, block Y2 does not Granger-cause block Y1. That is, Y1 is block-exogenous with respect to Y2.
- The test is usually a Wald or likelihood-ratio (LR) test, compared with a χ^2 distribution whose degrees of freedom equal the number of restrictions.

The key lesson is always to ask "what else belongs in the system?". Including the right variables can make false links disappear.

6. Application in Finance: Lead–Lag Analysis

Used honestly as a forecasting and screening tool rather than a causal claim, Granger causality shows which series carries information earlier than another. This is the essence of lead–lag analysis.

Typical lead–lag patterns

- Futures prices often lead spot prices (lower trading costs mean information is reflected there first).
- Liquid large-cap stocks can lead related small-cap stocks.
- Some macroeconomic releases lead sector returns.

Look first with cross-correlation

The cross-correlation function (CCF) shifts one series by k periods and measures its correlation with the other.

$$\rho_{XY}(k) = \mathrm{Corr}(X_{t+k},\, Y_t)\,, \qquad \text{peak at } \; k < 0 \;\Rightarrow \text{X leads Y}$$

In the lecture figure, the correlation peaked near 0.9 around k = −3. That means X's value three periods earlier matches Y's current value best. The CCF shows the leading relationship visually, and the Granger test confirms it statistically.

The efficient-market view

- If X reliably forecast Y's **returns**, traders would exploit it, and their trading would erase the pattern (Week 1, the efficient market hypothesis).
- Tradable, robust Granger causality in returns is therefore rare, and fragile when found.
- Where it does appear is at high frequency (the seconds to minutes it takes information to spread across assets), in volume, volatility and fundamentals rather than returns, and in transient lead–lag effects that vanish once they become known.

How to use it

- Use it as a screening tool for choosing candidate predictors, and always validate out of sample.
- Use it as a description of information flow, not as a guarantee of profit.
- Use it together with economic reasoning. Do not rely on a single test result.

7. Correlation, Granger Causality and True Causation

Granger causality sits between correlation and causation. It adds time ordering and forecasting to correlation, but it gives no grounds for intervention.

Property	Correlation	Granger causality	True causation
Uses time ordering	No	Yes	Yes

Property	Correlation	Granger causality	True causation
Forecasting power	Weak, static	Present by definition	Present
Rules out confounders	No	No	Yes (with a design)
Interpretation allowed	Association	Forecasting	Intervention

What a significant result does and does not tell you

What it tells you	What it does not tell you
In this sample, X's past contributes to forecasting Y	X is the cause of Y in a mechanical sense
There is a directional, time-ordered statistical link	The link will hold out of sample and in the future
X is a candidate input for a model that forecasts Y	No omitted variable explains the link

Proper use

Do	Don't
Use it as a forecasting and screening tool	Read it as cause and effect
Test both directions and control for related variables	Ignore confounders and market expectations
Trust it only after out-of-sample validation	Assume in-sample results will persist

8. Common Misconceptions and Self-Check

Common misconception	Reality
Granger causality proves that X causes Y	It proves only forecasting information, not a mechanism (Section 4)
Testing one direction is enough	Both directions must be tested to tell one-way, feedback and independence apart (Section 3)
Significant in sample means money to be made	Out-of-sample validation is needed, and efficient markets erase easy opportunities (Section 6)

Three questions before moving on

1. **The test shows that X Granger-causes Y. Can you conclude that X is the cause of Y?** No. Only a forecasting relationship has been established.
2. **Both X→Y and Y→X are significant. What is this called?** Bidirectional feedback.
3. **A significant link disappeared once variable Z was added. What happened?** Z was a confounder that moved X and Y together.

Next: Part 3

We resolve the problem of shocks being entangled in the same period (instantaneous causality, Section 5) through orthogonalization, then trace the path along which shocks spread through the system (IRF) and the sources of forecast error variance (FEVD).